

P2E: EVOLVING EVALUATION SYSTEMS FOR OPEN-ENDED GENERATION

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ABSTRACT

Human preferences specify desired orderings of open-ended outputs without identifying what an evaluator must observe or how it should interpret that evidence. We present P2E, Preference-Guided Evaluator Evolution, a framework for learning executable evaluation procedures under domain-specific human standards. Two data-construction settings reuse existing preferences or collect new preferences for tasks without suitable evaluation data. Both address the same question of evolving an evaluator from a task and its human feedback. P2E maps existing metrics, judges, and tools to quality dimensions and organizes structured feedback around semantically distinct disagreements. This evidence grounds hypotheses about missing observations, unused evidence, or inappropriate criteria. Within P2E, ERA evolves artifact representations, rubrics, evaluator modules, tools, evidence routing, and fusion while holding the base model fixed. Mechanism hypotheses guide connected changes, whereas complete programs are the units of selection. This separation permits evidence-guided continuation without equating an unsuccessful implementation with a refuted hypothesis. The research scope spans structured visual artifacts, images and 3D, text generation, and automated research. We specify matched-budget studies of feedback acquisition and evolution, followed by independent tests of alignment and selection, critic-guided refinement, and planned reward-guided training. Label provenance, annotation quality, and outcomes are reported separately by construction setting. Data construction does not establish the value of disagreement sampling, which requires a matched-budget random-sampling control. Quantitative displays in this draft use simulated data, not empirical results.

1 INTRODUCTION

An appealing image may omit a requested object. A fluent summary may remove a qualification that changes the source’s meaning. A research proposal may offer an interesting hypothesis without an experiment capable of testing it. These are evaluation problems in open-ended generation: multiple outputs are reasonable, quality has several dimensions, and no single deterministic verifier defines the complete ordering. Existing metrics and model-based judges provide useful but different evidence about that ordering (Fabbri et al., 2021; Xu et al., 2023; Ghosh et al., 2023; Liu et al., 2023).

The central difficulty is that a preference error does not identify its cause. A summary evaluator might never inspect the source qualification, observe it without passing it to the judge, or judge faithfulness correctly but give fluency too much weight. The same wrong ordering is compatible with each mechanism. Adding another score does not resolve this ambiguity when sources measure different constructs or share a blind spot. Work on model-based judging and human-authored criteria motivates examining what supports a judgment (Zheng et al., 2023; Shankar et al., 2024). The resulting question is: *given a task and its human feedback, can we evolve an evaluator that better follows that task’s evaluation standard?*

We introduce P2E, Preference-Guided Evaluator Evolution, for domains with open-ended outputs and an existing evaluation ecosystem. Figure 1 presents the complete framework. P2E maps avail-

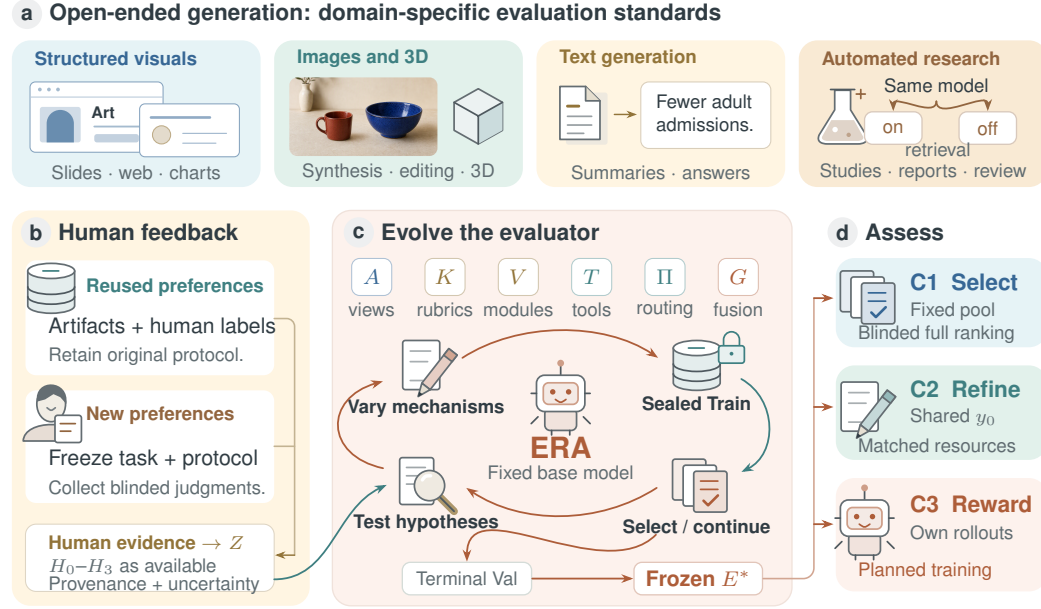


Figure 1: P2E evolves evaluation systems using human feedback. (a) Four artifact families retain distinct evaluation standards. (b) Reused and newly collected preferences enter a common evidence interface; their provenance and annotation quality remain separate. (c) ERA evolves a six-component program with a fixed base model. (d) C1 selection, C2 refinement, and planned C3 reward training. Artifacts and feedback are constructed illustrations; the photograph is generated with GPT Image 2. Arrows denote information flow, not measured improvements.

able signals to quality dimensions before exposing human labels. Suitable existing preferences can be reused; tasks without them require new human judgments. These are two data-construction settings for the same algorithmic problem, not different definitions of the method. Human evidence has two roles: it helps locate evaluation gaps and supplies the standard against which candidate mechanisms are selected. Within P2E, ERA is our method for evidence-guided evaluator evolution. With the base model fixed, it evolves how an executable program represents artifacts, acquires observations, and turns them into judgments. The object of learning is the evaluation procedure itself.

Two hypotheses motivate the design. First, disagreements between reliable, nonredundant signals may make a fixed annotation budget more informative than uniform sampling. Second, failures that depend on acquiring, routing, and interpreting evidence may require coordinated evolution of those components. A new observation cannot affect a judgment unless it reaches a module able to interpret it. ERA therefore organizes variation around mechanism hypotheses while selecting complete programs. A failed implementation need not refute its hypothesis, and continuation need not promote the current program. These distinctions motivate matched-budget comparisons with independent proposals and component-restricted search.

A common evolution procedure is instantiated separately for each evaluation standard. We distinguish agreement and selection on fixed candidates (C1), refinement through a frozen critic (C2), and training under a frozen reward (C3). Success at one level does not establish success at the next. This paper develops the feedback-mining formulation, ERA evolution procedure, and controlled studies needed to test these hypotheses in both data-construction settings. The ability to build a new preference corpus and the value of a particular sampling policy require different evidence.

2 RELATED WORK

Evaluators and feedback acquisition. G-Eval and MT-Bench study model-based judgment and human agreement (Liu et al., 2023; Zheng et al., 2023). Prometheus, Prometheus 2, and Self-Taught

Evaluators improve judges through rubric-conditioned or synthetic supervision (Kim et al., 2024a;b; Wang et al., 2024). EvalGen emphasizes aligning the criteria themselves with human preferences, while RewardBench and JudgeBench assess evaluators on challenging comparisons (Shankar et al., 2024; Lambert et al., 2024; Tan et al., 2025). Disagreement-based acquisition has a long history in query-by-committee (Seung et al., 1992). P2E applies this principle to an ecosystem whose signals need not measure the same construct. Its semantic mapping distinguishes same-dimension conflict from trade-offs and records source dependence. Dimension-wise human judgments can then locate discrepancies that an overall label alone does not resolve.

Optimizing language-model programs. DSPy exposes compositional pipelines to optimization (Khattab et al., 2024). ProTeGi, OPRO, TextGrad, and GEPA use model-generated feedback or proposals to improve instructions and programs (Pryzant et al., 2023; Yang et al., 2024; Yuksekgonul et al., 2025; Agrawal et al., 2026). ADAS, AFlow, and the Darwin Gödel Machine investigate automated construction or evolution of agentic systems (Hu et al., 2025; Zhang et al., 2025b;a). Their search spaces provide relevant controls, rather than a basis for assuming that prior methods cannot change structure. ERA specializes the representation and evidence interface to evaluation, linking ranking failures to observations, criteria, and synthesis while separating an implementation outcome from its underlying direction hypothesis.

Domain quality and downstream use. SummEval and BERTScore address summarization evaluation and text similarity (Fabbri et al., 2021; Zhang et al., 2020); ImageReward, Pick-a-Pic, and GenEval capture different aspects of image preference and alignment (Xu et al., 2023; Kirstain et al., 2023; Ghosh et al., 2023). These signals are useful evidence, not interchangeable definitions of quality. ReAct, Reflexion, and Self-Refine connect observations, reflection, and iterative improvement (Yao et al., 2023; Shinn et al., 2023; Madaan et al., 2023). Human-feedback summarization, instruction tuning, and direct preference optimization connect evaluation to model learning (Stiennon et al., 2020; Ouyang et al., 2022; Rafailov et al., 2023). P2E therefore separates fixed-pool evaluation, critic-guided refinement, and reward-guided training as distinct tests of an evolved evaluator.

3 SCOPE AND PROBLEM FORMULATION

A *domain* is a collection of tasks sharing core quality dimensions, their priorities, the overall decision rule, and the human annotation protocol. An artifact type does not define a domain: screenshot-conditioned webpage reconstruction and specification-conditioned webpage generation require different standards. A change of source distribution that preserves the standard is instead a within-domain distribution shift. Appendix A enumerates the full research scope of 22 domains in four families, without assuming that all are ready for experiments. Closed tasks whose complete quality criterion is an executable postcondition fall outside this scope.

Existing human preferences are not a domain-admission requirement. Their availability determines the missing data work, not the algorithm’s definition. An evaluation ecosystem can exist without a suitable preference dataset. We distinguish *reused preferences*, which test improvement on established benchmarks, from *new preferences*, which test applicability when matching human judgments must first be collected. One domain can support both settings.

Let x be a task with permitted context, y a candidate output, and $\mathcal{Q}_D = \{q_1, \dots, q_m\}$ the quality dimensions for domain D . A human comparison has the core form

$$h = (x, y_a, y_b, q, p, a, \nu), \quad (1)$$

where q is a dimension or overall quality, p is a preference, tie, or abstention, a records annotator agreement, and ν records provenance. Feedback has four levels: H_0 gives overall preference; H_1 gives dimension-wise preference; H_2 identifies error type, severity, and location; H_3 optionally supplies a free-form rationale. H_1 is central to the dimension-guided design. An H_0 -only corpus remains usable, but it supplies neither human dimensional judgments nor human explanations.

The learned object is an executable evaluation program

$$E_\theta = (A, K, V, T, \Pi, G), \quad (s_\theta, \tau_\theta) = E_\theta(x, y), \quad (2)$$

comprising artifact adapters A , dimension schema and rubrics K , evaluator modules V , tools T , evidence routing Π , and fusion G . The score s_θ is accompanied by an execution trace τ_θ . The base

model is fixed outside θ . At deployment, the evaluator receives one candidate and its permitted evidence; opponents, human labels, generator identity, split labels, and mining strata are excluded. We seek agreement with unseen human preferences at an explicitly reported evaluation cost, rather than a single judge shared across all domains.

4 EVALUATOR-GAP-DRIVEN PREFERENCE MINING

4.1 AN EVALUATION SEMANTIC SPACE

Feedback acquisition begins with a versioned inventory of evaluation knowledge, not with a convenient labeled dataset. For each domain, we record dimension definitions, human protocols, rubrics, published metrics, learned scorers, LLM/VLM judges, and external analysis tools. Each source has an input contract, provenance, known limitations, and an execution status. A coverage audit checks the resulting dimension–source matrix. Every core dimension needs executable evidence; conflict mining ordinarily needs two sources with different mechanisms. Several scorers sharing a model or training labels do not establish independent coverage.

The unit of comparison is an *output signal*, rather than an evaluator name. We freeze a many-to-many mapping M_{jq} from signal j to dimension q using its definition and measurement procedure before reading preference labels. A mixed score may connect to several dimensions, but remains one shared signal. Uninterpretable outputs are not assigned dimensions from post hoc correlations. Existing executable metrics can enter the candidate tool library. Metrics reserved for final independent assessment are frozen separately and never enter tools, routing, critique, or reward.

4.2 TWO ROUTES TO HUMAN FEEDBACK

Reusing a preference requires an aligned task, readable same-input candidates, and a judgment of the intended quality construct. Rankings, pointwise ratings, and error annotations need a validated adapter with conversion provenance. An input, reference, or automatic score alone is not human feedback.

When matching preferences are absent, we freeze the task and annotation protocol before collecting them. Existing candidates are retained when suitable; only missing candidate groups require fresh generation. Source-disjoint partitions are fixed before label-blind acquisition. New judgments are independent of old winners, model scores, and method identities. Their recency does not make them more reliable. Both settings report the annotator pool, supervision levels, agreement, ties, abstentions, and unresolved cases, with unavailable information marked as unknown (Appendix B). A protocol change defines a different measurement condition, not evidence that old labels were wrong.

4.3 SEMANTIC DISAGREEMENT AND STRUCTURED ANNOTATION

Raw scores on incompatible scales are not directly subtracted. For a candidate pair $r = (x, y_a, y_b)$, each signal supplies a direction $u_j(r) \in \{-1, 0, +1, \perp\}$, with a normalized margin, repeated-run variance, confidence, and availability. Zero denotes a tie and $\perp$ unavailable or abstaining evidence. A small random anchor estimates reliability and human disagreement before targeted sampling. For the reliable, nonabstaining signal pairs $\mathcal{P}_q(r)$ mapped to the same dimension, a conflict score is

$$c_q(r) = \frac{\sum_{(j,k) \in \mathcal{P}_q(r)} w_{jkq} \mathbb{I}[u_j(r) \neq u_k(r)]}{\sum_{(j,k) \in \mathcal{P}_q(r)} w_{jkq}}, \quad (3)$$

where nonnegative weights reflect anchor reliability and cap the influence of dependent sources. An empty denominator gives no conflict estimate. This formulation leaves the calibration and dependence rule to be fixed before the sampling comparison; disagreement is not itself a correctness label.

We sample three separately recorded semantic strata. *Within-dimension disagreement* occurs when reliable signals disagree about the same quality criterion. *Overall-dimension disagreement* indicates a possible mismatch in priorities or aggregation. *Cross-dimensional trade-offs* identify cases where different criteria favor different candidates; they are not contradictions between measurements of

the same construct. Model differences, repeated-run instability, and small margins further describe each stratum without replacing its semantic definition. Candidate groups cover generators, decoding choices, quality ranges, and failure types. After anchor calibration, the remaining pool’s labels stay hidden until the sampling policy is fixed. Only evolution data are conflict-sampled; final evaluation uses prespecified random or stratified sampling.

4.4 THE ADAPTATION EVIDENCE TABLE

Annotation targets overall and dimensional preferences, with localized diagnostics on a conflict or failure subset. All candidates derived from one input remain grouped. The adaptation evidence table $Z(r, q)$ joins artifact provenance, human judgments and agreement, every source’s direction and uncertainty, evaluator decisions, selected routes, tool evidence, and execution cost. Both construction settings supply this interface to ERA, while retaining their label origins and annotation protocols. Neither model-generated explanations nor labels derived from a different quality construct are promoted to human H_1 or H_2 .

5 ERA: EVIDENCE-GUIDED EVALUATOR EVOLUTION

5.1 AN EXECUTABLE EVALUATOR

ERA treats the evaluator’s observation and judgment mechanisms as the objects of evolution. Artifact adapters determine what the program can inspect, from rendered views and source text to spatial structure or interaction traces. Routing selects tools and sends their outputs to evaluator modules with explicit responsibilities. The modules apply the rubric and return findings with provenance and uncertainty; fusion turns these findings into a score and critique (Figure 4). These components are complementary: an available tool changes the judgment only if its evidence is acquired, routed, and interpreted. Existing metrics can serve as tools when their input contracts are satisfied, without requiring every tool to run on every artifact.

Fusion may use a declared numerical rule or semantic synthesis. Semantic synthesis reads specialist findings and their anchored quality scales, groups overlapping or conflicting factors, and locates overall quality on its own scale. It does not average raw scores with different meanings. References to findings must be accounted for, although correct reference bookkeeping does not establish semantic correctness. Appendix E illustrates this distinction.

5.2 MECHANISM HYPOTHESES GUIDE PROGRAM VARIATION

An evolution direction is a hypothesis about why the current program disagrees with human judgment. A summary evaluator, for example, may check individual facts without comparing their scope against the source. The direction predicts that making source qualifications available to the faithfulness module will change its ordering on scope errors while preserving judgments on faithful summaries. Original training artifacts and repeated traces ground the hypothesis, its competing explanations, target and protection cases, and editable components.

The hypothesis is the unit of exploration; the complete program is the unit of selection. Each elementary edit changes an attributable component of (A, K, V, T, Π, G) . A mechanism that needs a new tool, a route, and a consumer change can be instantiated as one connected candidate with a fixed parent. The complete candidate is measured once as a system; its outcome is not credited separately to every edit. Edit-size and component constraints bound variation, but do not guarantee a small semantic change. Modules may be split, merged, or removed as their responsibilities evolve.

5.3 SELECTION AND EVIDENCE-GUIDED CONTINUATION

ERA separates proposal evidence from candidate selection. The proposal subset $\mathcal{D}_{\text{prop}}$ exposes authorized training artifacts, available human feedback, and versioned traces. A source-disjoint, sealed training subset $\mathcal{D}_{\text{dev}}$ supplies aggregate development measurements only. Local checks on $\mathcal{D}_{\text{prop}}$ assess predictions and regressions without estimating population performance. A frozen candidate is also measured on the common $\mathcal{D}_{\text{dev}}$ cohort. Validation $\mathcal{D}_{\text{val}}$ is terminal; Test/OOD remain locked until the method and analysis are frozen (Appendix C).

Search retains both a best development program and a working program for the active direction. New directions begin at the best program. Continuing a direction may retain a fully measured intermediate program when new evidence supports further evolution and the protection conditions hold. Development promotion requires improved agreement, complete execution, nondecreasing coverage, and a prespecified stratum-regression guard. It is a selection rule, not a significance test.

Allocation can request evidence, continue, pause, or switch direction. Resource exhaustion is distinct from refutation. A direction is refuted only within a declared scope when evidence contradicts its prediction and relevant alternatives, including missing observations, unused tools, and incorrect fusion, have been examined. Failure of one candidate or a nonsignificant interval is insufficient. On normal completion, an eligible shortlist is frozen from training evidence before terminal validation. System confirmation uses fresh repeated scoring; component attribution is a separate controlled comparison. Neither substitutes for independent Test/OOD evaluation. Appendix D gives Algorithm 1 and the attribution contract.

5.4 POINTWISE MEASUREMENT

Each artifact receives $R \geq 3$ complete scores under a fixed protocol. Aggregation precedes comparison:

$$\bar{s}_\theta(x, y) = \text{median}_{r=1, \dots, R} s_\theta^{(r)}(x, y), \quad z_i = \mathbb{I}[\bar{s}_\theta(x_i, y_i^+) > \bar{s}_\theta(x_i, y_i^-)]. \quad (4)$$

Here y_i^+ is the human-preferred artifact, a designation never exposed to the scorer. Predicted ties are incorrect on strict-preference pairs; human ties require a separate prespecified policy. This is not majority voting over pairwise decisions. Missing executions retain their denominators and are reported through coverage and missing-outcome bounds (Appendix F).

6 EXPERIMENTAL DESIGN

Core comparisons, ablations, and analysis address distinct evaluation questions. Domains enter confirmatory experiments after a readiness audit; models, data, human protocols, budgets, and primary contrasts are frozen before final evaluation. Every result is indexed by domain, data-construction setting, and ID/OOD source, with its provenance and annotation quality. Reuse tests improvement with existing labels; collection tests applicability when labels must be acquired. Neither estimates the causal effect of label origin.

6.1 CORE RESULTS

C1: evaluator alignment and selection. Frozen P2E evaluators are compared with the best single metric, a metric ensemble, a strong static judge, and a static tool-augmented judge. Prompt-optimization and program-search controls use declared versions and comparable edit interfaces. Matched comparisons share the base model, human supervision, inputs, candidate pools, and scoring protocol. Search cost and deployment cost are reported separately.

For each test input, all evaluators receive the same candidate group, with a default design of $N = 4$. Complete human rankings support global pairwise agreement, top-region agreement involving the human top-two candidates, Best-of- N selection, and selection regret. When only ranks are available, normalized regret is $(\text{rank}(\hat{y}) - 1)/(N - 1)$ for the selected output $\hat{y}$. A pair-only dataset supports pairwise evaluation, not an N -way selection claim. Core H_1 , worst-stratum performance, coverage, variance, and cost accompany overall agreement.

ID and OOD results are reported separately. OOD changes dataset or source distribution while preserving the evaluation standard. A difficult subset alone is a difficulty test, not an OOD source. Final sampling is independent of mining conflicts. All splitting and statistical resampling preserve original input/source groups. Paired effects use grouped 95% intervals. Existing annotations are reused only when their construct and candidate groups match the endpoint.

C2: critic-guided refinement. For each unseen input, a frozen generator produces one fresh output y_0 . The seed critic and P2E critic each inspect that same output and permitted evidence, then return the same critique format. A common generator performs one refinement step under identical prompts, decoding, tool access, and token budgets. Refinement validation fixes these settings before

ID/OOD generation. New blinded human rankings assess y_0 , Seed-refined, and P2E-refined. The primary contrast is P2E-refined versus Seed-refined, accompanied by improvement and regression rates against y_0 , core dimensional judgments, and reserved external metrics. Labels on pre-existing outputs cannot assess these new outputs.

C3: reward-guided training. The planned study compares Base, Seed-GRPO, and P2E-GRPO from one checkpoint, using LoRA and group relative policy optimization (Hu et al., 2022; Shao et al., 2024). Training arms match prompts, optimizer settings, updates, policy compute, and reward-call budgets; scoring cost is reported separately. Rewards are frozen before training. Rollouts are sampled online by each arm’s own evolving policy and are not shared. New blinded H_0/H_1 and reserved metrics assess fresh ID/OOD outputs. Selection and reward-hacking safeguards appear in Appendix H. C3 remains a design-only study here; no parameter training or training benefit is reported.

6.2 ABLATION STUDIES

The ablations separate annotation choices from evaluator and search changes (Table 6). Mining strategies are compared at equal human budgets $N_H \in \{20, 50, 100, 200\}$, measured in original-input annotation units. Random and disagreement sampling use the same pool, feedback granularity, initial program, and evolution budget; reliability anchors are charged to every arm. Sampling rules are frozen before non-anchor labels are revealed, including in public-corpus replay. The endpoint is post-evolution alignment on an independently sampled cohort, not errors found during acquisition. Corpus construction alone cannot establish sampling efficiency. The value of H_1/H_2 requires a separate same-pair granularity comparison reporting both unit counts and annotation time. Component controls preserve unaffected interfaces and report invalid dependencies rather than reverting the whole program to its seed.

6.3 ANALYSIS AND STUDIES

Trajectory analysis records directions, implementation attempts, resource use, and evidence for scoped refutation. Component effects and remaining human-agreement gaps motivate tests for specification, observation, or headroom limits; strata within a domain may differ. Feedback analysis relates held-out changes to annotation units, time, and granularity. Adapter analysis compares direct judgments with labels derived from rankings, scores, or errors on the same subset. Repeated scoring and independent search runs remain distinct sources of variation.

7 QUANTITATIVE EVALUATION: SIMULATED DISPLAYS

All numerical displays in this section use *simulated data*, not measured evaluator performance or collected human judgments. They instantiate the reporting protocol while empirical evaluation remains pending. The generator, data, and limitations are documented in Appendix J. No ordering among the simulated methods constitutes evidence for either research hypothesis.

Alignment under distinct standards. The relevant comparison is evaluator evolution within a task, not one data source against another. Table 1 separates the two construction settings and the ID/OOD endpoints. Figure 2a–b retains domain-level paired effects instead of reducing heterogeneous standards to one average. Each effect compares P2E with the static tool-augmented seed on their jointly completed input groups. A change in pairwise agreement also leaves open which candidates are selected; the rank-composition display in Figure 6 makes that distinction explicit.

Acquisition as a separate test. A preference corpus can support evolution without establishing that its sampling policy was efficient. Figure 3a–b specifies the separate comparison against random sampling at equal human budgets. The endpoint is held-out agreement after evolution, not disagreement density in the acquired batch. The same reliability anchor is charged to all policies. Annotation granularity is held fixed in this comparison and varied separately in Figure 7b.

Table 1: **Simulated C1 reporting example; not experimental results.** Each setting averages four equally weighted toy cohorts. Pair = pairwise agreement; Top = top-region agreement; Bo4 = Best-of-4 accuracy (all in %); Reg = normalized rank regret (lower is better). Cohort assignments do not assert actual label availability. Coverage and domain summaries accompany the reproducible fixtures.

Evaluator	ID test				OOD test			
	Pair	Top	Bo4	Reg	Pair	Top	Bo4	Reg
SIMULATED Reused preferences								
Single metric	67.2	68.3	46.5	0.30	64.1	64.9	42.8	0.33
Metric ensemble	70.1	70.9	49.6	0.27	67.9	68.8	46.9	0.29
Static judge	72.3	73.4	54.3	0.23	70.6	72.5	52.2	0.25
Static tools	73.8	74.9	55.9	0.23	71.5	73.5	55.6	0.23
Prompt optimization	74.1	75.7	55.3	0.22	72.4	73.0	55.3	0.24
Program search	74.8	76.2	57.6	0.21	72.6	74.1	55.5	0.23
P2E (ERA)	77.7	79.1	62.7	0.18	74.2	75.7	59.0	0.21
SIMULATED New preferences								
Single metric	65.8	66.8	44.0	0.31	62.2	62.8	37.9	0.36
Metric ensemble	68.9	70.0	50.2	0.28	65.6	66.3	42.3	0.33
Static judge	71.5	72.3	55.4	0.23	68.0	69.1	48.3	0.29
Static tools	72.9	73.8	55.0	0.23	69.9	71.3	51.7	0.27
Prompt optimization	73.8	75.1	55.1	0.23	69.5	71.2	50.8	0.26
Program search	75.3	76.3	56.5	0.22	71.4	73.1	56.5	0.23
P2E (ERA)	75.7	76.9	58.8	0.21	72.4	73.7	55.3	0.23

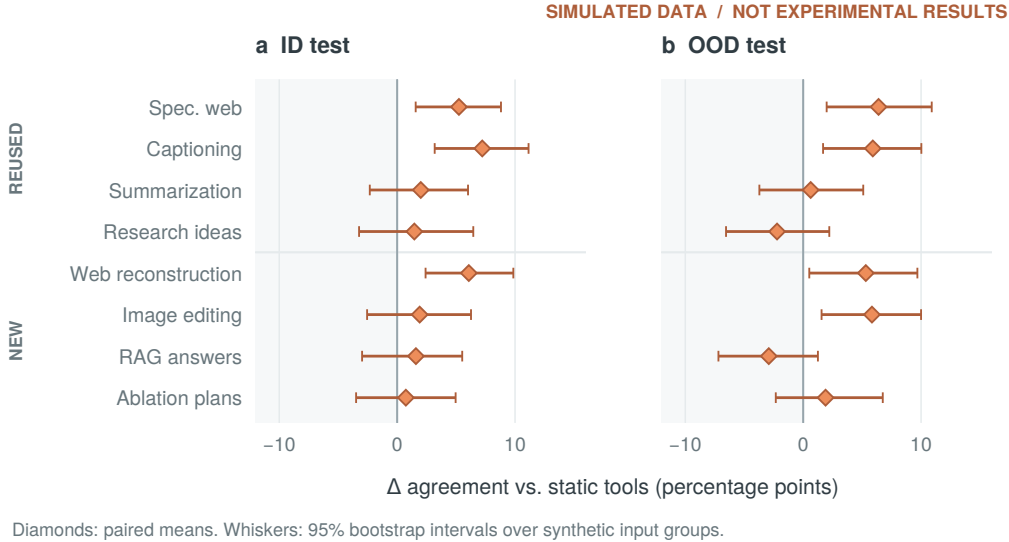


Figure 2: **Simulated domain-level alignment; not experimental results.** Panels (a) and (b) compare P2E with the static tool-augmented seed on ID and OOD inputs. Diamonds are paired mean differences; whiskers are 95% bootstrap intervals over jointly completed synthetic input groups. Zero denotes no difference. The two row groups indicate illustrative feedback-construction settings, not permanent domain categories or measured data readiness.

Feedback quality and unresolved judgments. Label origin does not determine reliability. Figure 5a–c separates valid-pair agreement, requested-rating abstention, and group consensus, preserving their distinct denominators. Unreported reliability in reused sources is marked unknown. C2 distributions and deployment-cost displays appear in Figures 8 and 9; C3 remains entirely unmeasured.

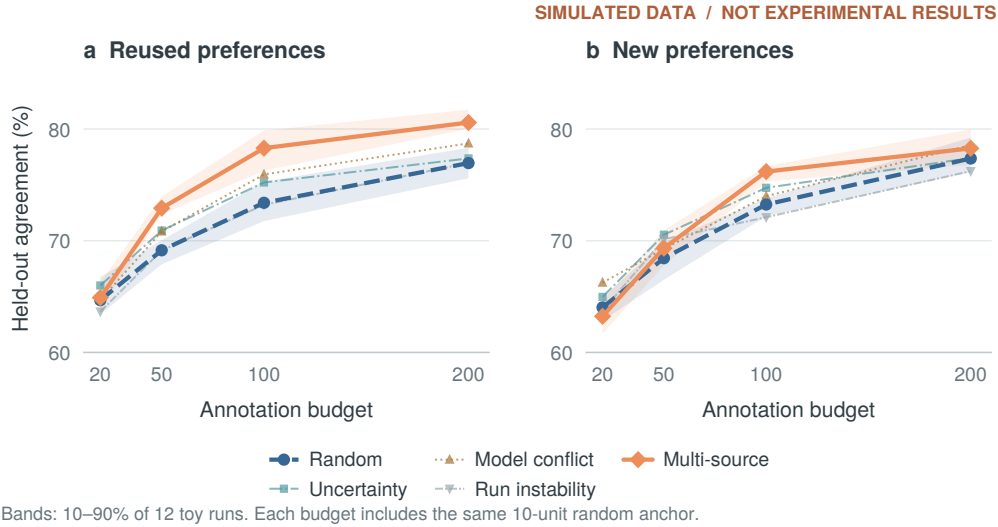


Figure 3: **Simulated acquisition curves; not experimental results.** Panels (a) and (b) compare five policies under reused and newly collected preferences. Lines average 12 toy runs; bands mark the 10th–90th percentiles for random and multi-source sampling, not confidence intervals. Budgets count original-input annotation units, including a shared 10-unit random anchor.

8 DISCUSSION

An ordering constrains evaluation without identifying which evidence was missing or how it should be weighted. ERA therefore treats observation, routing, and interpretation as objects of evolution. Structured feedback informs mechanism hypotheses; complete-program comparison determines which hypotheses yield an evaluator worth retaining.

Label availability changes the cost of posing this problem, not its meaning. Neither reused nor new preferences automatically validate the other; agreement and abstention expose uncertainty in each human endpoint. Data collection makes evolution possible, whereas a matched random-sampling comparison tests acquisition efficiency.

C1, C2, and C3 require separate evidence: correct rankings need not yield actionable critiques, and fixed-pool agreement need not survive policy optimization.

8.1 LIMITATIONS AND FUTURE WORK

First, simulated displays support no effectiveness claim and must be replaced with frozen empirical cohorts. Second, dimensional judgments require additional annotation and construct validation. Third, shared signal bias and interacting components require dependence audits and controlled attribution. Finally, pointwise scoring assumes a useful within-domain ordering, while repeated development can overfit aggregate feedback. Independent ID/OOD and generation-level studies must test evaluation gains and downstream utility; transfer across standards remains a separate question.

9 CONCLUSION

P2E treats evaluator evolution as learning to acquire and interpret evidence under a human evaluation standard. Reused and newly collected preferences support the same learning problem, with separate provenance and quality audits. ERA explores mechanism hypotheses and selects complete programs with a fixed base model. Across four domain families, the study design separates alignment, critic-guided refinement, and planned reward-guided training to ask when better judgment improves generation. Simulated displays illustrate these comparisons without resolving them.

AI USE STATEMENT

Generative AI assisted manuscript drafting, literature organization, and the preparation of editable LaTeX conceptual diagrams and plotting code. GPT Image 2 generated the photograph in Figure 1a through scene generation and a controlled edit. The quantitative displays use explicitly simulated data, generated offline for reviewing the reporting design. They are not observed cases, human annotations, model executions, or performance measurements. No empirical results or parameter-training runs were produced for this draft. Bibliographic identities were checked against primary metadata sources. Author review, responsibility for the final content, and the final disclosure of research-stage AI use remain to be completed before submission.

ETHICS STATEMENT

Human preferences can encode annotator and model biases rather than a universal definition of quality. Domain-specific criteria and uncertainty should remain visible, especially for automated research judgments. Any new annotation study must document consent, compensation, privacy protection, and the applicable institutional review determination before data collection. Artifact licenses and source permissions must be checked independently of technical accessibility. Generated code and evidence tools require restricted execution and controlled external access. Reward-guided optimization requires checks for exploitation of the evaluator and degradation outside its favored criteria. No private artifacts, identifiable annotation records, or experimental traces are included in this working manuscript. Study-specific approvals and release decisions are pending.

REPRODUCIBILITY STATEMENT

Sections 3–6 define semantic mappings, feedback, the evaluator program, evolution, and controlled evaluation. Appendices B–F specify annotation, access boundaries, the algorithm, and measurement. Appendices G–K specify ablations, planned training, reporting outputs, simulated fixtures, and the experiment manifest. Appendix J identifies the offline generator and provenance for the numerical displays. The current manuscript contains no empirical results to reproduce. Before submission, the authors must supply a versioned, appropriately anonymized implementation and permitted data-processing and analysis artifacts with completed manifests. Their availability is not asserted here.

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A DOMAIN PANORAMA AND EVALUATION STANDARDS

The research scope is organized by evaluation standard, not by file format. Table 2 retains all 22 candidate domains. Its dimensions are a starting taxonomy to be refined from domain evaluation knowledge before selecting datasets. Each instantiated domain must additionally fix priorities, decision rules, and an annotation protocol. Inclusion in this panorama does not establish data readiness, an implemented adapter, or empirical transfer. Figure 1a gives representative artifacts; the table makes their differing evaluation criteria explicit.

Table 2: Complete domain panorama. The four families organize the scope; every row requires its own standard and readiness audit. These are study candidates, not 22 completed evaluations.

ID	Domain	Core quality considerations
Structured visual artifacts		
TD-SV1	Brief-conditioned single-slide generation	Brief coverage, hierarchy, readability, composition, and visual consistency.
TD-SV2	Screenshot-conditioned webpage reconstruction	Reference content, spatial arrangement, typography, and visual fidelity.
TD-SV3	Data-to-visualization generation	Data fidelity, valid encodings, readability, and communicative usefulness.
TD-SV4	Specification-conditioned webpage generation	Requirement satisfaction, functional behavior, usability, and visual organization.
TD-SV5	Source-grounded slide-deck generation	Source faithfulness, narrative, coverage, slide readability, and deck consistency.
TD-SV6	Topic-conditioned slide-deck generation	Relevance, factual support, narrative completeness, audience fit, and presentation quality.
Image generation and understanding		
TD-IG1	Detailed image-caption generation	Grounding, coverage, relation accuracy, specificity, and absence of hallucination.
TD-IG2	Text-to-image generation	Prompt alignment, object and relation accuracy, visual coherence, and preference.
TD-IG3	Single-source instruction-based image editing	Instruction fulfillment, preservation of unrequested content, and image quality.
TD-IG4	LR-conditioned image super-resolution	Reference fidelity when available, perceptual detail, artifact avoidance, and plausible texture.
TD-IG5	Text-to-3D object generation	Prompt correspondence, geometry, appearance, and multi-view consistency.
TD-IG6	Image-to-3D object generation	Source-view fidelity, plausible unseen geometry, appearance, and multi-view consistency.
Text generation		
TD-TG1	Source-grounded document summarization	Faithfulness, relevance, coverage, coherence, and concision.
TD-TG2	RAG-supported long-form answer generation	Answer correctness, evidence grounding, attribution, completeness, and organization.
TD-TG3	Machine translation	Meaning preservation, omissions and additions, fluency, terminology, and style.
TD-TG4	Instruction-constraint-following response generation	Constraint satisfaction, substantive correctness, relevance, and response quality.
TD-TG5	Open-ended dialogue-response generation	Contextual appropriateness, helpfulness, consistency, safety, and conversational quality.
Automated research		
TD-AR1	Research-idea generation	Novelty relative to evidence, significance, feasibility, and testability.
TD-AR2	ML/NLP ablation-plan generation	Hypothesis identification, controlled contrasts, confound management, and feasibility.
TD-AR3	Table-to-insight report generation	Numerical faithfulness, appropriate inference, uncertainty, and relevance.

Table 2 continued

ID	Domain	Core quality considerations
TD-AR4	Related-work generation	Coverage, citation correctness, faithful comparison, and synthesis.
TD-AR5	Peer-review critique generation	Technical accuracy, evidence, significance assessment, fairness, and actionable criticism.

Representative resources include PPTAgent and PresentBench for presentations, Design2Code for screenshot-conditioned webpages, SummEval for summarization, and image-preference and alignment resources (Zheng et al., 2025; Chen et al., 2026; Si et al., 2025; Fabbri et al., 2021; Kirstain et al., 2023; Xu et al., 2023; Ghosh et al., 2023). Their outputs, references, metrics, and annotations have different evidential roles; a resource is not automatically a source of the required human preference. A benchmark collection spanning several standards remains a dataset collection, not an additional domain. Research execution with a complete executable postcondition is excluded, while ideation, study design, interpretation, writing, and critique remain open-ended evaluation problems.

Dataset roles. Dev supports acquisition and evaluator development, ID tests unseen inputs from the development source distribution, and OOD changes source distribution while preserving the standard. External datasets provide separately locked assessment only and do not determine semantic mappings or acquisition. Harder examples from the same source are a difficulty stratum, not a new OOD domain. A domain selected for confirmatory execution must pass the audit below; selection does not remove unselected domains from the research scope.

B FEEDBACK PROVENANCE, ANNOTATION, AND READINESS

The two construction settings in Table 3 share the same evaluator-evolution problem. They differ in the work needed to obtain matched human feedback. A domain may use both settings. Direct, derived, and materials-only provenance describe what a resource supplies; they are not additional domain classes or alternative algorithms.

Inputs, references, editable artifacts, and automatic metrics without matched human judgments are materials only. They may support candidate construction or independent assessment, but do not supply human supervision. When H_0 is absent, we preserve the original task and evaluation standard, separate source groups, recover readable candidates or generate them with frozen generators, and collect the required feedback. Preparation can precede annotation; final human-alignment claims cannot precede matched human assessment.

Annotation units and levels. One annotation unit is an original input and its anonymized candidate group, including permitted references. Pairs derived from that group remain dependent. The core tuple in Equation 1 is retained together with dataset and license, source grouping, generator and prompt version, collection time, annotator pool, agreement, and conversion provenance. Generator identity is recorded for audit but hidden from annotators and the deployed scorer.

The planned evolution protocol collects H_0 and core H_1 , with H_2 on conflict and failure subsets. H_3 is optional. Two independent annotators with adjudication may support development; final ID/OOD assessment targets three independent annotators and blinded complete rankings. Core dimensions and the tie policy are fixed in advance. A model or an author decision is not substituted for a missing independent final annotation. An H_0 -only instantiation must be reported as such; the framework’s richer interface does not establish that a run actually received H_1 or H_2 .

Agreement, abstention, and reannotation. New labels are not assumed to be more reliable than reused labels. Report the agreement statistic, its valid-response denominator, and the number of raters per group. Abstention uses all requested ratings as its denominator, including those excluded from valid-pair agreement. Group outcomes distinguish decisive preference, tie, no consensus, and insufficient votes. The simulated audit in Figure 5a–c illustrates these separate quantities. Individual disagreements are retained rather than filtered to match an old winner.

Table 3: Two routes to human feedback. Report each source’s protocol, agreement, abstention, and available supervision separately. An adapter changes representation, not the quality construct or the provenance of a judgment.

Setting	Label origin and protocol	Permitted supervision
Reused preferences	Original human pairs/rankings on readable same-input candidates; retain the source protocol and document any conversion from rankings, scores, or errors	H_0 only for overall judgments; H_1/H_2 only when actually annotated. Unknown source agreement or abstention remains unknown
New preferences	Freeze the task, visible evidence, dimensions, ties, and abstention before collecting independent blind judgments on existing or freshly generated candidates	Collect the levels required by the study; preserve individual votes and unresolved disagreements. Fresh C2/C3 outputs require new endpoint judgments

Table 4: Readiness criteria before a domain enters confirmatory evaluation. These are audit requirements, not completed coverage scores.

Audit object	Required record and exclusion condition
Evaluation standard	Open-endedness, dimensions and hierarchy, priorities, decision rule, and human protocol. Exclude tasks fully decided by an executable postcondition.
Knowledge provenance	Benchmarks, expert guidance, rubrics, and methods with versions, dates, input assumptions, known biases, and availability.
Source coverage	Dimension $\times$ source-type matrix: deterministic, learned metric, LLM/VLM judge, external analysis tool. Each core dimension needs executable evidence.
Source independence	For conflict mining, ordinarily two mechanisms per dimension. Record shared models, training labels, metrics, and mixed signals; multiple names are not independent votes.
Executable artifacts	Recoverable input/candidate groups, readable artifacts, references, permissions, and source fingerprints. Missing assets do not become scorer disagreement.
Conflict pool	Three semantic strata, availability, margins, repeat variance, anchor reliability, dependence caps, and uncovered strata. Noise alone is not a gap.
Feedback fidelity	Direct versus derived provenance, agreement, mapping to $H_0/H_1/H_2$, adapter checks on a common subset, and additional annotation needs.
Source isolation	Original-input deduplication, Dev/ID/OOD/External roles, contamination review, and a pre-search External Metric Reserve.

Annotators do not see model scores, method identity, or earlier winners. A changed rubric or visible evidence defines a changed protocol; resulting differences do not by themselves establish errors in the older labels.

Label-blind acquisition. Freeze domain knowledge, signal definitions, semantic mappings, and the calibration recipe before accessing preference labels. Select a random anchor, reveal only its authorized training judgments to estimate reliability, then freeze the targeted sampling policy before opening labels in the remaining pool. Charge anchor annotations to every strategy’s human budget. An existing labeled dataset must simulate the same access order. The final cohort is sampled independently of conflicts and never contributes to reliability calibration.

Source and tool status. Registry entries distinguish executable, planned, and blocked sources (Table 4). A publication or code link without compatible inputs, weights, or dependencies is not an executed signal. Available tools begin as candidates; a route must explicitly activate them. Source independence is assessed at the signal level, including model, training-label, and metric dependencies. Model-only comparisons hold rubric, evidence, context, and reasoning budget fixed. Replacing both the scoring model and its evidence-collection agent instead compares full systems. Deterministic or fixed learned metrics are not duplicated for every judge model.

Table 5: Information available to evolution and assessment. Every invocation of the deployed scorer remains pointwise and label-blind, including on Train.

Partition	Information released	Permitted decision
Proposal Train	Authorized artifacts, feedback, and traces	Mechanism hypotheses and candidate evolution
Sealed Train	Prespecified aggregates only	Development promotion, continuation, and allocation
Terminal Val	Frozen-shortlist measurements	Prespecified terminal selection; no new mechanism edits
Test/OOD	Locked until method and analysis freeze	Independent final reporting
Metric reserve	No access during evolution	Final assessment only; no tools, critique, reward, or selection

C DATA ACCESS AND INDEPENDENT ASSESSMENT

Table 5 separates proposal evidence, adaptive development measurements, and terminal assessment. The scorer always receives the same pointwise input contract, regardless of the cohort in which it is invoked. Only the authorized diagnostic process sees human labels and pair identities.

Training and validation. Proposal-Train evidence may inform evolution directions, candidates, and allocation. Sealed Train reveals prespecified aggregate agreement, coverage, and stratum guards without examples, labels, or per-case errors. Aggregates remain adaptive development evidence, not a final performance estimate. The terminal shortlist and its order are fixed from training evidence before Val access, and every shortlisted outcome is reported. Repeated Val calls can assess stability but do not create an independent test cohort. Interrupted searches do not obtain a new terminal assessment merely because they stopped.

External metrics. The External Metric Reserve is frozen before search. Reserved metrics and their equivalent wrappers are excluded from the executable tool library, router, diagnostician, critique, reward, and checkpoint-selection criteria. If an endpoint was used for optimization, it loses independent-assessment status and must be replaced or explicitly reported as an optimization metric. This restriction does not forbid using other existing metrics as tools.

D ERA ALGORITHM AND ATTRIBUTION CONTRACT

Figure 4 separates the editable observation and judgment mechanisms. Algorithm 1 specifies their evolution and selection.

Hypothesis state. Each direction records a mechanism, its evidence, alternative explanations, a falsifiable prediction, protected cases, editable components, and the observations needed to assess the prediction. Local diagnostics are not a mandatory success gate for broader sealed-development measurement. Continuation retains a working program only when its measurement, new evidence, and protection conditions justify doing so. The best program is a separate selection state; retaining a working program does not redefine the best score.

Promotion and confirmation. Development promotion requires complete execution, a strict agreement gain, nondecreasing coverage, and a prespecified worst-stratum regression guard. Guard thresholds and tie rules are fixed before search. These are development selection rules rather than statistical significance claims. Normal completion is followed by fresh repeated terminal measurements of the frozen shortlist. Selection uncertainty and the number of assessed candidates are reported.

Attribution. An elementary mutation changes one attributable component. A connected patch may introduce a tool, a route, and a consumer change against one fixed parent. The bundle receives one system-level outcome, not three independent causal credits.

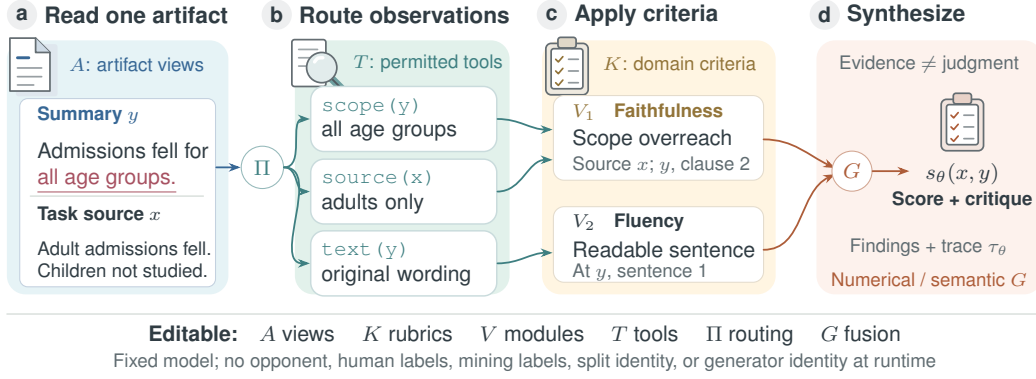


Figure 4: The six-component evaluation program. (a) Read one candidate and permitted task evidence. (b) Route tool observations. (c) Apply dimensional criteria. (d) Fuse sourced findings into a score and critique. The lower rail identifies the editable components. Human feedback supervises evolution, never the deployed scorer. Evidence and judgment remain separate; the summary and findings are a constructed example.

Algorithm 1 ERA within the P2E framework

Require: Seed E_0 , proposal evidence Z_{prop} , sealed development interface, fixed budgets and measurement rule

- 1: Measure E_0 on the common sealed Train cohort; set best program $B \leftarrow E_0$
- 2: Initialize direction history and allocation state
- 3: **while** search resources remain **do**
- 4: Choose to request evidence, start, continue, pause, or switch direction
- 5: **if** requesting evidence **then**
- 6: Acquire authorized proposal-Train evidence; charge its cost
- 7: **else if** starting or continuing a direction **then**
- 8: Set parent $P \leftarrow B$ for a new direction, or the eligible working program
- 9: Specify prediction, alternatives, target cases, and protection conditions
- 10: Construct a bounded atomic edit or connected patch $E' \leftarrow \text{patch}(P)$
- 11: Inspect local diagnostics; retain observations and failures
- 12: Measure frozen E' on the common sealed Train cohort
- 13: Record the full attempt, component changes, costs, and aggregate outcome
- 14: **if** measurement is complete and the development promotion rule holds **then**
- 15: $B \leftarrow E'$
- 16: **end if**
- 17: **if** measurement is complete and the continuation rule holds **then**
- 18: Retain E' as the direction's working program
- 19: **end if**
- 20: Update direction status without equating a failed candidate with refutation
- 21: **else**
- 22: Save the direction state and reallocate the remaining resources
- 23: **end if**
- 24: **end while**
- 25: On normal completion, freeze an eligible shortlist using training evidence only
- 26: Assess the shortlist on terminal Val under a prespecified selection rule
- 27: Freeze the final evaluator and analysis before independent Test/OOD assessment

Component attribution requires paired interventions with all other interfaces held fixed, including tool on/off, evolved versus seed tool, and adaptive versus static routing. If disabling a dependency makes a program invalid, report that limitation and use a compatible control where justified. System confirmation and attribution answer different questions.

A scoped refutation records the contradicted prediction and the evidence that rules out relevant implementation alternatives. Missing observations, an unselected tool, incorrect interpretation, or insufficient resources leave different conclusions. Neither one unsuccessful implementation nor a nonsignificant difference alone refutes the direction.

E SEMANTIC SYNTHESIS AND ANCHORED SCORING

Semantic synthesis is an optional fusion procedure, compared with a declared numerical rule. A specialist report contains a finding identifier, supporting evidence, a dimension, uncertainty, and an anchored quality judgment. The synthesizer groups findings into factors while preserving overlap and contradiction. Duplicate descriptions of the same defect should not become independent penalties, and a favorable style judgment should not silently cancel a factual error when the domain standard prioritizes faithfulness.

Let $a_k < a_{k+1}$ be adjacent declared anchors and $\lambda \in [0, 1]$ locate quality between them. The resulting scalar is

$$s = (1 - \lambda)a_k + \lambda a_{k+1}. \quad (5)$$

The anchors have domain-specific verbal meanings. Finding-level scores with different meanings are not averaged merely because they are numeric. Reference completeness, semantic correctness, and calibration are separate properties and require separate checks. A synthetic arithmetic example uses $a_k = 50$, $a_{k+1} = 75$, and $\lambda = 0.4$, giving $s = 60$. These numbers do not prescribe a universal scale or report an evaluator result.

F MEASUREMENT AND STATISTICAL ANALYSIS

Completeness and denominators. For a prespecified strict-preference cohort of N pairs, let $c_i = 1$ when both artifacts have all R valid complete scores. Let $m = \sum_i c_i$ and $k = \sum_{i:c_i=1} z_i$, with z_i from Equation 4. Report completed-pair accuracy k/m when $m > 0$, coverage m/N , and the missing-outcome interval

$$\left[\frac{k}{N}, \frac{k + N - m}{N} \right]. \quad (6)$$

This interval bounds full-cohort agreement under arbitrary missing outcomes; it is not a statistical confidence interval. A missing repeat is not replaced by the median of surviving calls. Completeness includes selected routes and tools, not only a final score. A valid observation that an artifact fails a quality check differs from an execution error.

Ranking and selection endpoints. For the default $N = 4$ design, human complete rankings support all strict pairs. Global agreement averages within input groups before aggregation across inputs. Top-region agreement uses prespecified strict comparisons involving at least one human top-two candidate. Best-of- N measures selection of a human top-ranked candidate; normalized rank regret is defined in Section 6.1. Human ties, evaluator ties, and incomplete rankings are governed by a frozen policy and reported separately. Pair-only data do not supply missing ranks or cardinal utility. Core H_1 agreement is reported by dimension before any macro average.

Randomness and repeated use. Freeze $R \geq 3$, score precision, retry policy, artifact reuse, and cache identity. Retries are outcome-blind and uniformly applied. Repeated scores are not independent task observations. Paired uncertainty uses original-input/source group resampling, preserving candidate groups and comparisons; the planned interval level is 95%. Report independent search seeds separately from scoring repeats, and distinguish within-run execution variation from between-search variation. Retain failed, rejected, incomplete, and interrupted attempts in the authorized analysis, without favorable resampling or denominator changes.

C2/C3 human outcomes. Report wins, losses, ties, invalid outputs, and major regressions per input group. A prespecified tie-aware preference probability may assign half credit to a tie, with its denominator and missingness bounds stated. Report C2 changes against the shared y_0 in addition to the critic contrast, and C3 changes against Base in addition to the reward contrast. Intervals and costs accompany effect sizes. Multiple domains, dimensions, and ablations require declared primary contrasts and a prespecified multiplicity policy. A search-selected development maximum is not substituted for any final endpoint.

Table 6: Planned ablation conditions and the inference each comparison supports. All conditions require matched evidence access and accounted computation. An oracle condition is diagnostic, not a deployable comparator.

Factor	Conditions	Endpoint or interpretation
Acquisition	Random; judge uncertainty; cross-model conflict; run instability; P2E semantic multi-source disagreement	Final held-out alignment at equal $N_H \in \{20, 50, 100, 200\}$; acquisition yield alone is insufficient.
Human detail	Same pairs with H_0 , $H_0 + H_1$, $H_0 + H_1 + H_2$, and $H_0 + H_1 + H_2 + H_3$	Marginal value of granularity; report annotation units and annotation time.
Editable scope	Full (A, K, V, T, Π, G) ; rubric/procedure only (K, V) ; fixed tools and static routes; routing only	Whether additional observations or structural changes are needed beyond criteria changes.
Evidence ceiling	Oracle evidence and/or oracle routing with access declared	Diagnostic headroom under privileged information; no deployment claim.
Search allocation	Independent one-shot proposals; history-only proposals; direction depth-first; direction search with pause/resume	Value of direction management at equal total search and measurement budgets.
Tool activation	Tool on/off with compatible consumers	Contribution of acquiring the observation, not the whole program bundle.
Tool implementation	Evolved tool versus seed tool	Contribution of changed measurement or evidence extraction.
Evidence route	Adaptive versus fixed routing with the same tool set	Contribution of which observations reach which modules.
Rubric and modules	Evolved versus fixed criteria/procedures	Contribution of judgment changes with observations held fixed.
Fusion	Declared numerical rule versus semantic synthesis	Contribution of resolving priorities, overlap, and conflicting findings.
Feedback adapter	Direct pairs versus derived ranking/pointwise/error labels on the same subset	Label fidelity and its downstream effect; keep original input dependence.

G FULL ABLATION MATRIX

The comparisons in Table 6 freeze the initial program, evidence access, edit interfaces, proposal opportunities, and measurement budgets. All diagnostic and validation calls are charged, including unsuccessful attempts. If method interfaces differ, report that difference instead of attributing a resource advantage to the search policy. Attribution uses paired original-input groups and pre-specified compatible component replacements. Observed interactions may limit a one-component interpretation.

H PLANNED C3 TRAINING PROTOCOL

C3 is specified to test reward usefulness independently of evaluator agreement and critic usefulness. It is not an executed experiment, and no parameter training is part of this manuscript preparation.

Under the controls in Table 7, the frozen reward produces scalar scores within each rollout group using the same normalization and advantage rule. Zero-variance groups, invalid artifacts, and execution failures require a common prespecified policy. Deterministic checks may enforce format or executability, but do not define the entire reward for open-ended quality.

The pilot uses development validation only. If its prespecified stability or reward-hacking gate fails, expansion stops for that domain and the failure is retained. Success elsewhere is not a substitute. Final sample sizes are determined from pilot variance and a declared minimum detectable effect before test generation. Test inputs and judgments do not participate in training, checkpoint selection, reward design, or failure-driven edits.

Table 7: Controls for the planned LoRA + GRPO comparison. Numerical hyperparameters, pilot thresholds, and final sample sizes remain to be frozen in the experiment manifest.

Control	Contract
Arms	Base is untrained; Seed-GRPO and P2E-GRPO start from the identical base checkpoint.
Training data	Same unlabeled training inputs and order; source-level deduplication and contamination audit.
Policy updates	Same LoRA structure, optimizer, learning rate, update count, group size, KL rule, checkpoint cadence, and policy-training compute.
Online samples	Each updated policy samples its own rollout groups. Do not share trajectories once policies diverge.
Reward version	Freeze program, prompt, tools, routing, aggregation, normalization, ties, abstention, and invalid-output policy before training.
Reward resources	Equal reward-call counts and token caps; report actual inference time, tool use, and monetary or compute cost separately.
Training validation	Input-disjoint development validation selects checkpoints and checks reward exploitation; final metrics and final human labels remain locked.
Pilot gate	Prespecified acceptable validity, reward variance, KL, length, stability, held-out quality, and reward-hacking diagnostics before expansion.
Final generation	Freeze checkpoints and decoding, then generate new Base, Seed-GRPO, and P2E-GRPO outputs on the same unseen ID/OOD inputs.
Final judgment	New blinded H_0 and core H_1 , reserved external metrics, invalid/major-regression rates, length and format shifts, and total cost.

Any deliberate sharing of locked input pools across C1/C2/C3 must be prespecified with an access audit; their newly generated outputs and judgments are never treated as reusable old labels.

I REQUIRED RESULT AND ANALYSIS OUTPUTS

Empirical plots for the outcomes in Table 8 require frozen reporting cohorts and analysis. Suitable outputs include human-budget curves, per-domain component effects, annotated search trajectories, and C2/C3 regression breakdowns. The numerical displays in Appendix J are reporting fixtures, not provisional substitutes for those measurements. Table 9 records the empirical status of the downstream studies independently of the simulated C2 illustration.

Table 8: Required empirical outputs beyond the simulated reporting examples. Every result requires authorized frozen data and a traceable analysis; neither conceptual illustrations nor simulated displays supply such evidence.

Output	Required summary	Interpretation limit
C1 alignment	Global/top-region agreement, per-dimension H_1 , worst stratum, ID/OOD, coverage, repeated-score variation	Pairwise improvement does not establish selection or generation gains.
Feedback quality	Label origin, source protocol, annotator pool, agreement, abstention, unresolved judgments, and annotation time per setting	Newly collected labels are not automatically more reliable.
C1 selection	Best-of- N , normalized rank regret, complete human rankings, grouped intervals	Rank regret is not cardinal utility; pair-only corpora are insufficient.
C2 refinement	New blinded contrast, changes against y_0 , major regressions, dimensional quality, external metrics	Fresh shared initial outputs and matched refinement are necessary.
C3 training	New blinded reward contrast, Base sanity check, external metrics, validity, reward-hacking diagnostics	Design-only until separately authorized training and final evaluation occur.
Efficiency	Human units and time, search calls/tokens/tool use, runtime, deployment and training costs	More compute or annotation detail is not a free method gain.
Trajectory	Directions, implementation attempts, continuation/pause, exhaustion/refutation, resource allocation	A schematic tree is not an optimization trace or evidence of efficiency.
Mechanism	Paired component effects, interactions, remaining gap to human agreement, scoped counterexamples	Assign limitation regimes only after evidence; do not generalize from one stratum.
Adapter fidelity	Direct/derived label agreement and downstream differences on the same subset	Annotation conversion cannot change the construct being evaluated.

Table 9: Empirical downstream status, separate from all simulated displays. Dashes denote unmeasured quantities, not zeros. C3 is a planned study only.

Study	Primary contrast	ID preference	OOD preference	Regressions
C2	P2E-refined vs. Seed-refined	—	—	—
C3	P2E-GRPO vs. Seed-GRPO	—	—	—

J SIMULATED REPORTING FIXTURES

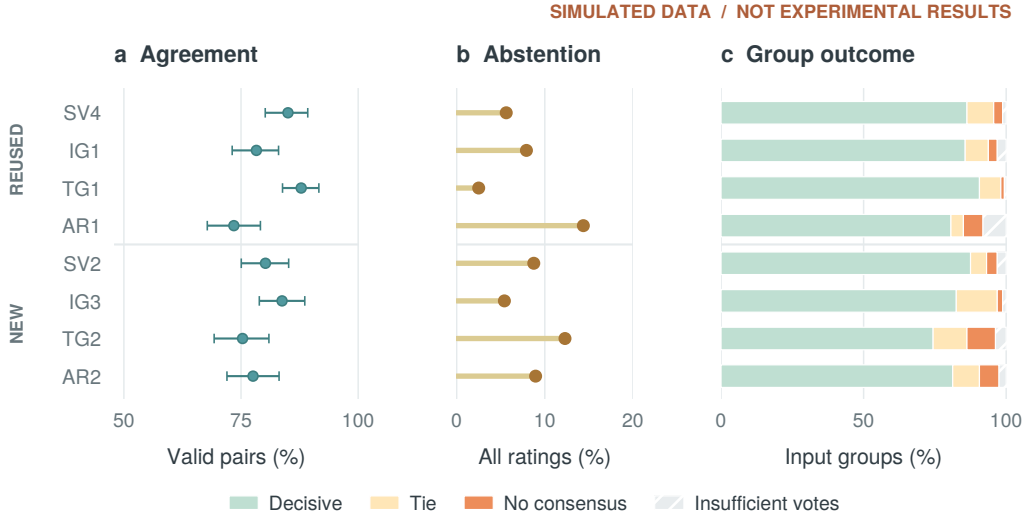
Purpose and limits. All quantitative figures and Table 1 are generated from explicitly simulated data for reviewing the reporting design. The generator does not execute ERA, run a model, consult benchmark outcomes, or recruit annotators. Its assigned noise, effects, costs, and cohort labels are not estimates of expected performance, a power analysis, or evidence for method ranking. Positive, near-zero, and negative contrasts are retained. None of these fixtures provides evidence for acquisition efficiency, component effectiveness, or human-label reliability. No C3 training outcomes are simulated.

Construction and uncertainty. Table 10 maps the eight toy cohorts to the full domain taxonomy. Gaussian latent utilities define synthetic C1 rankings. Noisy scores are repeated three times per artifact and reduced to medians before strict comparison. Incomplete groups remain in coverage denominators. The C1 table averages domain-level metrics equally within each setting. Its means use each method’s complete groups, whereas the paired differences in Figure 2 use the intersection of completed groups. Intervals use 1,000 input-group bootstrap replicates of these artificial observations. They quantify resampling variation in the fixtures, not uncertainty about ERA. The acquisition generator assigns budget responses for five policies and 12 toy runs; its bands are percentile ranges across runs, not bootstrap intervals.

Table 10: Illustrative cohorts in the simulated displays. Assignments are layout examples, not verified data availability or fixed domain categories. Each C1 domain/split contains 160 synthetic input groups and four candidates.

Family	Reused-preference example	New-preference example
Structured visuals	TD-SV4: specification-conditioned web	TD-SV2: webpage reconstruction
Images / 3D	TD-IG1: detailed image captioning	TD-IG3: instruction-based image editing
Text generation	TD-TG1: document summarization	TD-TG2: RAG-supported answers
Automated research	TD-AR1: research ideas	TD-AR2: ablation plans

The annotation fixture assigns three simulated raters per group. Candidate preferences are encoded as 0/1, a tie as 2, and abstention as -1 . Agreement is computed within each group from all available non-abstaining rater pairs, then averaged across eligible groups. Fewer than two valid votes is insufficient; three distinct valid votes yields no consensus. A two-of-three majority gives a decisive preference or a tie. These outcomes are not evidence about actual annotators or any public benchmark.

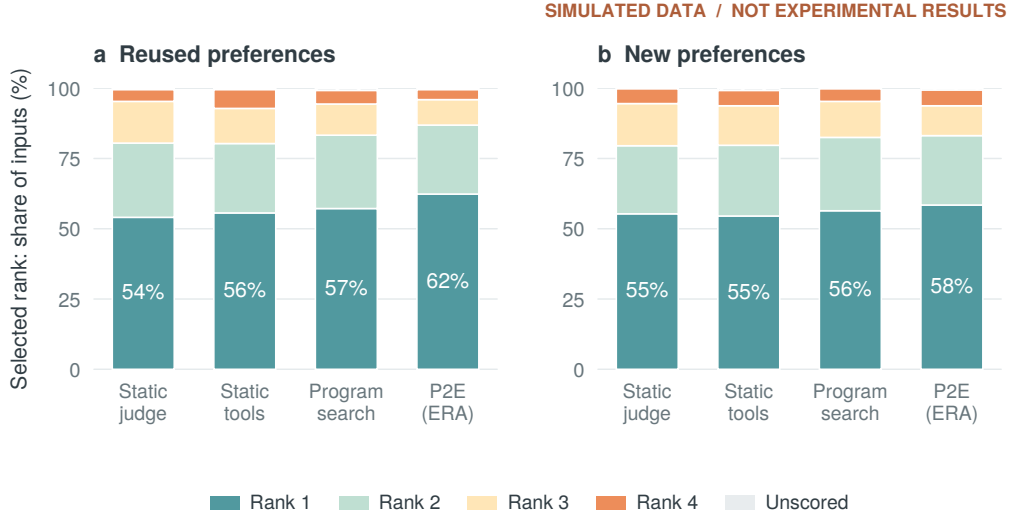


Three simulated raters per group. Agreement, abstention, and consensus use distinct denominators.

Figure 5: **Simulated annotation audit; no human study was conducted.** Panel (a) averages valid-pair agreement within input groups; whiskers are 95% group-bootstrap intervals. Panel (b) divides abstentions by all requested ratings. Panel (c) retains decisive preferences, ties, no consensus, and insufficient votes under a two-of-three rule. Domain codes map to Table 10. The settings share a schema, not an assumed reliability ordering.

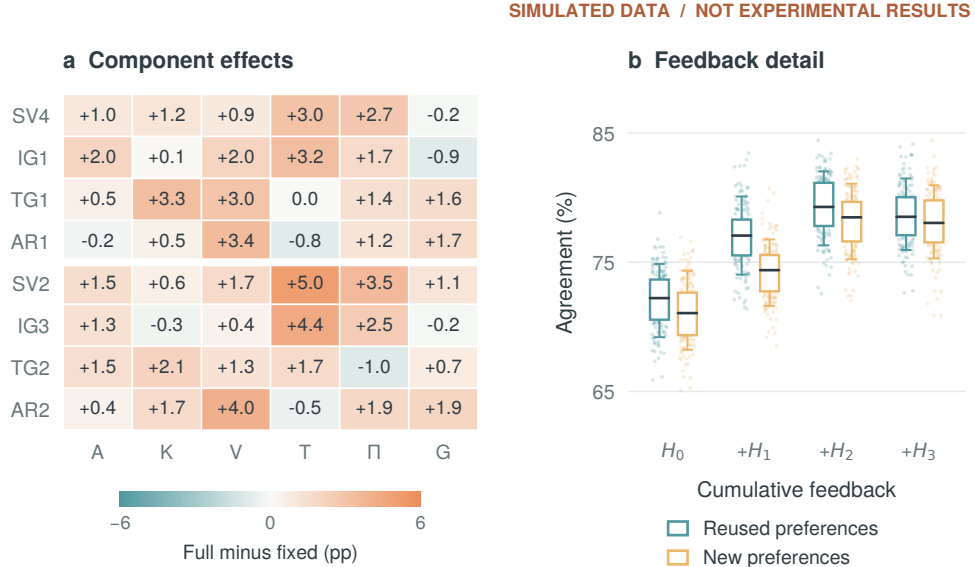
Selection beyond pairwise agreement. Figure 6a–b reports the entire selected-rank distribution. Rank 1 is the synthetic human optimum; unscored inputs remain visible. This full-input denominator differs from the completed-group Best-of-4 accuracy in Table 1. Both definitions are exported with their denominators so missingness cannot masquerade as better selection.

Mechanisms and feedback detail. The heatmap in Figure 7a specifies conditional contrasts between a full program and a program holding one component fixed. Such contrasts are not additive causal contributions. Panel (b) varies cumulative feedback on the same synthetic input groups. Its distributions describe assigned group-level values rather than independent search runs. Actual granularity studies must also account for annotation time and unequal label availability.



ID test; equal-sized toy cohorts. Rank 1 is best. Failed scoring remains in the denominator.

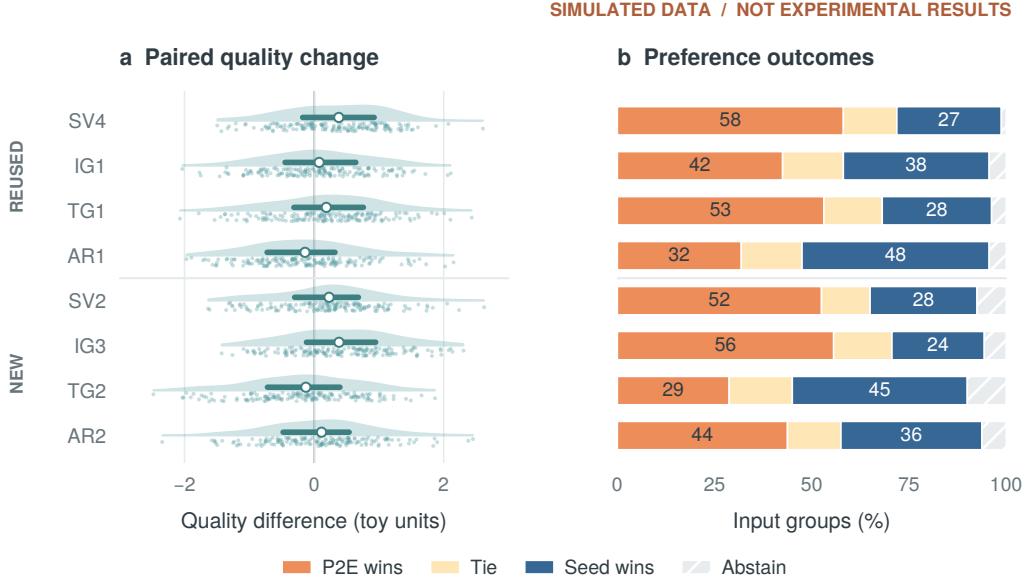
Figure 6: **Simulated C1 selection; not experimental results.** Panels (a) and (b) show selected ranks for four evaluator fixtures in the two construction settings, using ID inputs. Rank 1 is best. Labels give its share of all input groups, including unscored groups in the denominator. Colors encode ranks, not methods. Equal cohort sizes permit pooling within a setting; actual domain-level results must also be reported.



Synthetic paired effects are not additive. Boxes: quartiles; whiskers: 10–90%; dots: input groups.

Figure 7: **Simulated ablations; not experimental results.** Panel (a) gives mean full-minus-fixed agreement differences in percentage points for the six components (A, K, V, T, Π, G). Panel (b) compares cumulative H_0 – H_3 detail. Dots denote synthetic input groups; boxes show quartiles, central lines medians, and whiskers the 10th–90th percentiles. Colors in panel (a) encode signed differences; in panel (b) they identify construction settings. No displayed contrast establishes a component’s benefit.

Refinement and resource use. The C2 fixture constructs two noisy refinements from a shared latent y_0 . Figure 8a retains their paired quality differences and panel (b) converts those differences to wins, ties, and losses with an independent abstention indicator. The toy tie threshold is 0.15 latent units, not a proposed universal human threshold. Reused/new row labels describe feedback used for evaluator evolution; endpoint judgments on fresh C2 outputs must be new in both settings. Changes against y_0 and major-regression flags are retained in the data for the full reporting contract.



Difference = P2E-refined minus Seed-refined. All artifacts and judgments here are synthetic.

Figure 8: Simulated C2 outcomes; not experimental results. Panel (a) combines half-violin densities, all synthetic group values, median markers, and interquartile segments for P2E-refined minus Seed-refined quality. The horizontal scale is an arbitrary latent unit, not a human score. Panel (b) retains every win, tie, loss, and abstention under the toy rule. Both construction settings would require fresh blinded human endpoint judgments in an actual C2 study.

Figure 9a–b pairs agreement with assigned relative deployment costs, keeping the two settings separate. Dashed paths connect nondominated toy points only. A real resource comparison must report search and annotation costs separately from deployment cost; the artificial costs here cannot establish practical efficiency.

Reproduction and replacement. The offline script `scripts/render_simulated_results.py`, seeded with 20260916, produces the CSV fixtures, vector figures, numerical table, and a manifest of hashes. Every data row carries a `SIMULATED` provenance field, and every chart discloses its status within the image. The committed assets compile without model access. Before a results-bearing submission, replace the fixtures with authorized measurements, regenerate captions and uncertainty, and audit all claims against the actual evidence. Removing a disclosure without replacing its data is not a valid conversion to results.

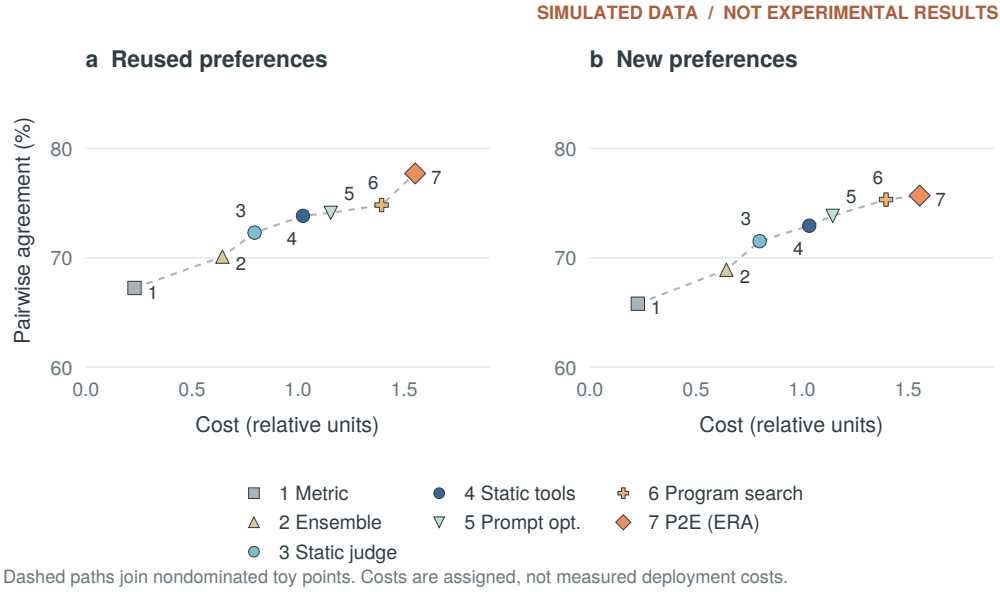


Figure 9: **Simulated quality–cost trade-offs; not measured costs.** Panels (a) and (b) give equal-weight means across four ID cohorts per setting. Markers identify evaluator fixtures and dashed paths join their nondominated points. Cost units are assigned to the toy generator, not timed API calls or measured compute. The plot does not establish an efficiency frontier for ERA.

K EXPERIMENT MANIFEST

Table 11: Information required before confirmatory evaluation. Unset values are intentional placeholders, not inferred development defaults.

Item	Required specification
Domain and sources	Versioned taxonomy, decision rule, readiness audit, dataset versions, permissions, input counts, candidate groups: <i>TBD</i>
Supervision	Direct/derived provenance, H_0 – H_3 availability, core dimensions, annotator pool, agreement, time, tie policy: <i>TBD</i>
Partitioning	Original-input/source grouping, deduplication, split sizes, OOD definition, External role, contamination review: <i>TBD</i>
Mining	Source inventory, mappings, dependence groups, anchor size, calibration, strata, frozen policy, human-budget accounting: <i>TBD</i>
Models	Frozen scorer, proposer, tool-model, and generator identities, revisions, decoding, and permissible evidence: <i>TBD</i>
Program and search	Seed, editable components, edit bounds, allocation, development guard, protection criteria, shortlist rule: <i>TBD</i>
Measurement	Repeats $R \geq 3$, precision, completeness, retries, artifact reuse, cache identity, search seeds, stopping rule: <i>TBD</i>
Budgets	Human time, opportunities, model calls/tokens, tools, runtime, deployment cost, training compute, matching rule: <i>TBD</i>
Downstream controls	C1 pool size; C2 fresh generation and refinement; C3 reward, LoRA/GRPO settings, pilot gates and preflight: <i>TBD</i>
Analysis	Primary contrasts, grouped intervals, multiplicity, missingness, reserved metrics, minimum effects, final sample sizes: <i>TBD</i>

Before submission, complete Table 11, replace result placeholders with permitted source-traceable measurements, finalize the empirical scope without changing the method’s definition of a domain, and supply appropriately anonymized reproducibility artifacts. Unmeasured hypotheses remain hypotheses.